

df galois theory

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1 Basic Definitions

Claim: $\text{Aut}(K/F) \leq \text{Aut}(K)$.

Proof: $1 \in \text{Aut}(K/F)$. Let $f, g \in \text{Aut}(K/F)$. Then $fg^{-1} \in \text{Aut}(K)$ and for all $x \in F$, $fg^{-1}x = fx = x$, so $fg^{-1} \in \text{Aut}(K/F)$.

Claim: Let $\alpha \in K$ be algebraic over F with minimal polynomial f . Let $\sigma \in \text{Aut}(K/F)$. Then $\sigma\alpha$ is a root of f , ie σ permutes the roots of f .

Proof: Write $f = (x - \alpha) \cdot g$ for some $g \in K[x]$. Then

$$\sigma f = f \rightarrow (x - \sigma\alpha) \cdot \sigma g = f,$$

so $f(\sigma\alpha) = 0$, ie $\sigma\alpha$ is a root of f . Let S denote the set of roots of f . Then the restriction $\sigma' : S \rightarrow S$ is well-defined. Since σ is injective, so is σ' . Since S is finite, σ' is bijective.

Claim: Let $H \leq \text{Aut}(K)$. Then the collection of elements in K fixed by all elements of H is a subfield of K .

Proof: Let $\varphi \in H$. It suffices to show $\text{Fix } \varphi \leq K$. Indeed: $1 \in \text{Fix } \varphi$, if $x, y \in \text{Fix } \varphi$, then $\varphi(x - y) = \varphi(x) - \varphi(y) = x - y$, so $x - y \in \text{Fix } \varphi$. If $y \neq 0$, then $\varphi(xy^{-1}) = \varphi(x)\varphi(y)^{-1} = xy^{-1}$, so $xy^{-1} \in \text{Fix } \varphi$. Note H need not be a subgroup of $\text{Aut}(K)$.

Claim: Aut, Fix are contravariant maps.

Proof: Let $F_1 \leq F_2$ and $\varphi \in \text{Aut}(K/F_2)$. Then $\varphi(x) = x$ for all $x \in F_1$, so $\varphi \in \text{Aut}(K/F_1)$. Let $H_1 \leq H_2 \leq \text{Aut}(K)$ and $x \in \text{Fix } H_2$. Then $\varphi(x) = x$ for all $\varphi \in H_2$, so $x \in \text{Fix } H_1$.

Claim: Let $f \in F[x]$ and E/F be a splitting field of f . Let $\varphi : F \rightarrow F'$ be an isomorphism, E'/F' , so φ extends to some $\sigma : E \rightarrow E'$. Then the number of such σ is at most $[E : F]$.

Proof: I induct on $[E : F]$. If $[E : F] = 1$, then $E = F$, so $\sigma = \varphi$ is unique. Suppose $[E : F] > 1$, so $F < E$. Pick $\alpha \in E - F$, so α is a root of a nonlinear irreducible g . The splitting field of f over E equals the splitting field of f over $F(\alpha)$ contained in E .

Let $\sigma : E \rightarrow E'$ be an isomorphism and $g' = \sigma g = \varphi g$, which is independent of σ . Note that $\sigma\alpha$ is sent to a root of g' , say β . Then σ restricts to an isomorphism $\sigma' : F(\alpha) \rightarrow F'(\beta)$. There are at most $[F(\alpha) : F]$ such isomorphisms, which equals the number of roots of g . The number of extensions of σ' is at most $[E : F(\alpha)] < [E : F]$, so there are at most $[E : F(\alpha)] \cdot [F(\alpha) : F] = [E : F]$ extensions of φ .

If f is separable and the g 's are taken to be nonlinear irreducible factors of f (over the current basefield), then equality holds.

Let $K = \mathbb{Q}(\sqrt[3]{2}, \zeta)$ be the splitting field of $x^3 - 2$, where ζ is a primitive 3rd root of unity, so $K/\mathbb{Q}$ is Galois. Let $\varphi \in \text{Aut}(K/\mathbb{Q})$. Then $\varphi(\sqrt[3]{2})^3 = \sqrt[3]{2}, \zeta\sqrt[3]{2}, \zeta^2\sqrt[3]{2}$ ($\sqrt[3]{2}$ is sent by φ to another root of the irreducible $x^3 - 2$) and $\varphi(\zeta) = \zeta, \zeta^2$ (ζ is sent to a root of $x^2 + x + 1$). Since $[K : \mathbb{Q}] = 6$, each of these cases gives an automorphism of $K/\mathbb{Q}$. Let

$$\begin{cases} \sigma(\sqrt[3]{2}) = \zeta\sqrt[3]{2} \\ \sigma(\zeta) = \zeta \end{cases} \quad \begin{cases} \tau(\sqrt[3]{2}) = \sqrt[3]{2} \\ \tau(\zeta) = \zeta^2 \end{cases}$$

so $\sigma^3(\sqrt[3]{2}) = \sqrt[3]{2}$ and $\sigma^3(\zeta) = \zeta$, so $\sigma = 1$. Also, $\tau^2(\sqrt[3]{2}) = \sqrt[3]{2}$ and $\tau^2(\zeta) = \zeta$, so $\tau^2 = 1$. Finally,

$$\begin{aligned} \sigma\tau(\sqrt[3]{2}) &= \sigma(\sqrt[3]{2}) = \zeta\sqrt[3]{2} \\ \tau\sigma(\sqrt[3]{2}) &= \tau(\zeta\sqrt[3]{2}) = \zeta^2\sqrt[3]{2}, \end{aligned}$$

so $\text{Gal}(K/\mathbb{Q})$ is nonabelian and therefore isomorphic to S_3 .

Claim: Let F be a field with $\text{char } F \neq 2$ and let K/F such that $[K : F] = 2$. Then $K = F(\sqrt{D})$ for some $D \in F$ not a square.

Proof: Pick $\alpha \in K - F$. Then $m_{\alpha, F}$ is quadratic, say $m_{\alpha, F} = x^2 + ax + b$. By the Quadratic formula, we have $\alpha = \frac{-a \pm \sqrt{a^2 - 4b}}{2}$. Let $D = a^2 - 4b$. If D is a square, then $\alpha \in F$, absurd. Then $K = F(\alpha) = F(\sqrt{D})$.

Claim: Let K/F with $\text{char } F \neq 2$. Then K/F is Galois.

Proof: We have $K = F(\sqrt{D})$, where D is not a square in F . Consider $f = x^2 - D$, which has no roots in F . Further, f splits over K as $f = (x - \sqrt{D})(x + \sqrt{D})$, so K is the splitting field of f and therefore Galois.

Note the inclusions $\mathbb{Q} \rightarrow \mathbb{Q}(\sqrt{2}) \rightarrow \mathbb{Q}(\sqrt[4]{2})$, each of which is quadratic over $\mathbb{Q}$ of characteristic not equal to 2, so the composition of Galois extensions need not be Galois.

Compute the splitting field of $x^8 - 2$ and its degree over $\mathbb{Q}$. I claim that $\mathbb{Q}(\sqrt[8]{2}, i)$ is the splitting field of $x^8 - 2$.

Proof: Let $K \leq \mathbb{C}$ be such that $f = x^8 - 2$ splits. Then $\sqrt[8]{2}, i\sqrt[8]{2} \in K$, so $i \in K$. I claim that f splits over $K = \mathbb{Q}(\sqrt[8]{2}, i)$. Indeed, the roots of $x^8 - 2$ are $\zeta^k \sqrt[8]{2}$, where ζ is a primitive 8th root of unity, $k \in \mathbb{Z}$. We compute $\zeta = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{\sqrt{2} + i\sqrt{2}}{2} = \frac{\sqrt{2}}{2}(1 + i) \in K$, so K is indeed splitting.

I claim that $[K : \mathbb{Q}] = 16$. Note that $[\mathbb{Q}(\sqrt[8]{2}) : \mathbb{Q}] = 8$ since $x^8 - 2$ is irreducible by Eisenstein. $[\mathbb{Q}(i) : \mathbb{Q}] = 2$ as usual. We have $[K : \mathbb{Q}] \leq [\mathbb{Q}(\sqrt[8]{2}) : \mathbb{Q}] \cdot [\mathbb{Q}(i) : \mathbb{Q}] = 16$. Sps FTSOC $[K : \mathbb{Q}] \neq 16$. Since $\mathbb{Q}(\sqrt[8]{2}) \leq K$, we have $[\mathbb{Q}(\sqrt[8]{2}) : \mathbb{Q}]$ divides $[K : \mathbb{Q}]$, so $[K : \mathbb{Q}] = [\mathbb{Q}(\sqrt[8]{2} : \mathbb{Q})] = 8$, so $[K : \mathbb{Q}(\sqrt[8]{2})] = 1 \rightarrow K = \mathbb{Q}(\sqrt[8]{2}) \rightarrow K \leq \mathbb{R}$, absurd.

$\mathbb{F}_{p^n}$ is the splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$, so $\mathbb{F}_{p^n}/\mathbb{F}_p$ is Galois. We have $[\mathbb{F}_{p^n} : \mathbb{F}_p] = n$, so $|\text{Aut}(\mathbb{F}_{p^n})| = n$. Let $\sigma_p \in \text{Aut}(\mathbb{F}_{p^n})$ be the Frobenius automorphism. We compute $\sigma_p^n(x) = x^{p^n} = 1x$, so $\sigma_p^n = 1$. Suppose FTSOC $\sigma_p^k = 1$ for some $0 \leq k < n$. Then $x^{p^k} = x$ for all $x \in \mathbb{F}_{p^n}$, so the polynomial $x^{p^k} - x$ has at least $p^n > p^k$ roots, absurd. Hence, $\text{Aut}(\mathbb{F}_{p^n}) \cong \mathbb{Z}_n = \langle \sigma_p \rangle$.

Consider $x^2 - t$ over $\mathbb{F}_2(t)$. Sps $p(x)^2 = tq(x)^2$ for some nonzero $p, q \in \mathbb{F}_2[t]$. Then the LHS has even degree, while the RHS has odd degree, absurd. Hence, $x^2 - t$ is irreducible. Note over $\mathbb{F}_2(t)(x)$, $x^2 - t = (x + t)(x - t) = (x - t)^2$, so $x^2 - t$ is inseparable.

Claim: $\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3})$ are not isomorphic.

Proof: Sps $\varphi : \mathbb{Q}(\sqrt{2}) \rightarrow \mathbb{Q}(\sqrt{3})$ is an isomorphism. Then $\varphi(\sqrt{2})^2 = 2$, so for some $a, b \in \mathbb{Q}$, we have $(a + b\sqrt{3})^2 = 2 \rightarrow (a^2 + 3b^2) + 2ab\sqrt{3} = 2 \rightarrow a^2 + 3b^2 = 2, 2ab = 0$. If either a, b are zero, we have a contradiction.

Claim: Let $K = F(\alpha_1, \alpha_2, \dots, \alpha_n)$. Suppose $\sigma(\alpha_i) = \tau(\alpha_i)$ for all $i \in [n]$. Then $\sigma = \tau$.

Proof: Let $\omega = \tau^{-1}\sigma$, so $\omega\alpha_i = \alpha_i$ for all $i \in [n]$. Since $\alpha_i \in K^\omega$ for all $i \in [n]$, we have $F(\alpha_1, \alpha_2, \dots, \alpha_n) \leq K^\omega$, so $K^\omega = K$, ie $\omega x = x$ for all $x \in K$.

Let K/F such that $G \leq \text{Aut}(K/F)$ has generators $\{\sigma_i\}_{i \in [n]}$. Suppose E/F is fixed by $\{\sigma_i\}$. Then E/F is fixed by G .

Let $x \in E$. We have $\sigma_i x = x$ for all $i \in [n]$, so $\sigma_i \in G_x$. Since G_x is a group, we have $G \leq G_x$, so $x \in K^G$, so $E \leq K^G$.

Compute $\text{Aut}(\mathbb{R}/\mathbb{Q})$.

Let $\sigma \in \text{Aut}(\mathbb{R}/\mathbb{Q})$, so $\sigma(xy) = \sigma(x)\sigma(y)$ for all $x, y \in \mathbb{R}$. In particular, for $x > 0$, we have $\sigma(x) = \sigma(\sqrt{x})^2 > 0$ since $\sigma(\sqrt{x}) \neq 0$ (since σ is injective). If $x < y$, then $\sigma(x) < \sigma(x) + \sigma(y - x) = \sigma(y)$, so σ is increasing.

Suppose FTSOC $\sigma(r) \neq r$ for some $r \in \mathbb{R}$. Suppose $\sigma(r) > r$. Pick by density of $\mathbb{Q}$ in $\mathbb{R}$ $q \in \mathbb{Q}$ such that $r < q = \sigma(q) < \sigma(r)$, absurd. Suppose $\sigma(r) < r$. Pick $\sigma(r) < \sigma(q) = q < r$, absurd. Hence, $\sigma = 1$, so $\text{Aut}(\mathbb{R}/\mathbb{Q}) = 1$.

Define $\varphi_{a,b} : k[t] \rightarrow k[t]$ by $t \mapsto at + b$ and $\varphi_{a,b}$ fixes k . If $a \neq 0$, note that

$$\begin{aligned}\varphi_{a,b}\varphi_{a^{-1},-a^{-1}b}(t) &= \varphi_{a,b}(a^{-1}t - a^{-1}b) = a(a^{-1}t - a^{-1}b) + b = t \\ \varphi_{a^{-1},-a^{-1}b}\varphi_{a,b}(t) &= \varphi_{a^{-1},-a^{-1}b}(at + b) = a^{-1}(at + b) - a^{-1}b = t,\end{aligned}$$

so $\varphi_{a,b}, \varphi_{a^{-1},-a^{-1}b}$ are inverses.

Suppose $\varphi : k[t] \rightarrow k[t]$ is an automorphism fixing k . Let ψ be the inverse and $t \xrightarrow{\varphi} f(t), t \xrightarrow{\psi} g(t)$. Then $f(g(t)) = t$, so f, g are degree 1.

If $ad \neq bc$, notice that

$$-\frac{c}{ad-bc}(at+b) + \frac{a}{ad-bc}(ct+d) = 1,$$

so $\gcd(at+b, ct+d) = 1$. Suppose $p = \sum_{i=0}^n p_i x^i$ has the root $x = \frac{at+b}{ct+d}$. Then $\sum_{i=0}^n p_i (ct+d)^{n-i} (at+b)^i = 0 \rightarrow (ct+d)^n \mid \lambda_n (at+b)^n \rightarrow (ct+d) \mid \lambda_n \rightarrow c = 0$. Then from the original equation, we have $a = 0$, so $ad - bc = 0$, absurd.

If p were nonzero constant, then p also has no roots. Hence, we have an injective map $\varphi : k[t] \rightarrow k(t)$ by fixing k and having $t \mapsto \frac{at+b}{ct+d}$. This gives an injective map $\varphi : k(t) \rightarrow k(t)$. We can construct the reverse map $\psi : k(t) \rightarrow k(t)$ given by $t \mapsto \frac{-dt+b}{ct-a}$, which works since $(-d)(-a) - bc \neq 0$.

Now suppose $\varphi : k(t) \rightarrow k(t)$ is an automorphism fixing k . We have for some rational functions f, g that $f(g(t)) = t$.

2 Fundamental Theorem of Galois Theory

Claim: Let $\chi_1, \chi_2, \dots, \chi_n$ be distinct with values in L . Then $\chi_1, \chi_2, \dots, \chi_n$ are linearly independent over L .

Proof: Pick m to be the minimal number of linearly dependent χ_i so that $\sum_{i \in [m]} a_i \chi_i = 0$. Then for all $g \in G$, we have $\sum_{i \in [m]} a_i \chi_i(g) = 0$. Pick $g_0 \in G$ such that $\chi_1(g_0) \neq \chi_m(g_0)$. Then

$$\begin{aligned} \sum_{i \in [m]} a_i \chi_i(g_0 g) &= \sum_{i \in [m]} a_i \chi_i(g_0) \chi_i(g) = 0 \\ \chi_m(g_0) \sum_{i \in [m]} a_i \chi_i(g) &= \sum_{i \in [m]} a_i \chi_m(g_0) \chi_i(g) = 0 \\ \rightarrow a_1(\chi_1(g_0) - \chi_m(g_0)) \chi_1(g) + \sum_{2 \leq i \leq m-1} a_i(\chi_i(g_0) - \chi_m(g_0)) \chi_i(g) &= 0 \end{aligned}$$

Since $a_1(\chi_1(g_0) - \chi_m(g_0)) \neq 0$, there exist $m-1$ linearly dependent characters (some subset of $\{\chi_i : i \in [m-1]\}$), absurd.

Claim: Let E/F be finite. Then $|\text{Aut}(E/F)| \leq [E : F]$.

Proof: I induct on $[E : F]$. If $[E : F] = 1$, then $E = F$, so $\text{Aut}(E/F) = 1$. Let $[E : F] > 1$, so $F < E$. Pick $\alpha \in E - F$ of degree n over F . Suppose $\sigma \in \text{Aut}(E/F)$. Then σ sends α to another root β of $m_{\alpha, F}$. Then σ extends an isomorphism $\tau_{\alpha, \beta} : F(\alpha) \rightarrow F(\beta)$. Note that $[E : F(\alpha)] = [E : F(\beta)] = [E : F]/n$, so the induction hypothesis applies. There are at most $[E : F]/n \cdot n = [E : F]$ elements of $\text{Aut}(E/F)$.

Claim: Let $G = \{\sigma_1 = 1, \sigma_2, \dots, \sigma_n\} \leq \text{Aut}(K)$ and let $F = K^G$. Then $[K : F] = n = |G|$.

Proof: Suppose $\{\alpha_j\}_{j \in [m]}$ is a basis of K/F . Define $M = [\sigma_i(\alpha_j)]_{(i,j) \in [n] \times [m]}$. Suppose there exist not all zero $\lambda_i \in K$ such that $\sum_{i \in [n]} \lambda_i \sigma_i(\alpha_j) = 0$ for all $j \in [m]$. Equivalently, we have $\sum_{i \in [n]} \lambda_i \sigma_i = 0$, absurd. It follows $n \leq m$.

Now suppose $n < m$. There exist $n+1$ elements $\alpha_i \in K$ for $i \in [n+1]$ linearly independent over F . Consider the systems

$$\begin{aligned} \sigma_1(\alpha_1)x_1 + \sigma_1(\alpha_2)x_2 + \dots + \sigma_1(\alpha_{n+1})x_{n+1} &= 0 \\ \sigma_2(\alpha_1)x_1 + \sigma_2(\alpha_2)x_2 + \dots + \sigma_2(\alpha_{n+1})x_{n+1} &= 0 \\ &\vdots \\ \sigma_n(\alpha_1)x_1 + \sigma_n(\alpha_2)x_2 + \dots + \sigma_n(\alpha_{n+1})x_{n+1} &= 0 \end{aligned}$$

in $n+1$ variables and n equations. There exists a nontrivial solution $(\beta_i)_{i \in [n+1]}$ to these equations.

Let $(\beta_i)_{i \in [r]}$ be a solution with r equaling the number of nonzero β_i (after renaming). Since we can scale, WLOG $\beta_r = 1$.

We have the systems $\sum_{j=1}^{r-1} \sigma_i(\alpha_j)\beta_j + \sigma_i(\alpha_r) = 0$ for $i \in [n]$. Since the α_j are independent over F , we have $\beta_j \notin F$ for some $j \in [r-1]$. WLOG $\beta_1 \notin F$, so $\sigma_{k_0}(\beta_1) \neq \beta_1$ for some $k_0 \in [n]$ (by definition of F).

Applying σ_{k_0} to the equation, we have $\sum_{j=1}^{r-1} \sigma_{k_0}\sigma_i(\alpha_j)\sigma_{k_0}(\beta_j) + \sigma_{k_0}\sigma_i(\alpha_r) = 0$. Since G is a group, there exists a permutation $\pi : [n] \rightarrow [n]$ such that $\sigma_{k_0}\sigma_i = \sigma_{\pi(i)}$. Then we have $\sum_{j=1}^{r-1} \sigma_{\pi(i)}(\alpha_j)\sigma_{k_0}(\beta_j) + \sigma_{\pi(i)}(\alpha_r) = 0$. Then reindexing, we have $\sum_{j=1}^{r-1} \sigma_i(\alpha_j)\sigma_{k_0}(\beta_j) + \sigma_i(\alpha_r) = 0$, so subtracting gives $\sum_{j=1}^{r-1} \sigma_i(\alpha_j)(\beta_j - \sigma_{k_0}(\beta_j)) = 0$ (for all $i \in [n]$). Since $\beta_1 - \sigma_{k_0}(\beta_1) \neq 0$, we have a nontrivial solution in β_i with at most $r-1$ nonzero components, absurd.

Corollary: Let K/F be finite. Then $|\text{Aut}(K/F)| \leq [K : F]$ with equality iff F is the fixed field of $\text{Aut}(K/F)$. In other words, K/F is Galois iff F is the fixed field of $\text{Aut}(K/F)$.

Proof: I claim that $K^{\text{Aut}(K/F)} \geq F$. If $x \in F$ and $\sigma \in \text{Aut}(K/F)$, then $\sigma(x) = x$, so $x \in K^{\text{Aut}(K/F)}$. It follows $|\text{Aut}(K/F)| = [K : K^{\text{Aut}(K/F)}] \leq [K : F]$.

If $|\text{Aut}(K/F)| = [K : F]$, then $[K : F] = [K : K^{\text{Aut}(K/F)}]$, so $F = K^{\text{Aut}(K/F)}$.

Corollary: Let $G \leq \text{Aut}(K)$ be finite and $F = K^G$. Then $\text{Aut}(K/F) = G$.

Proof: We have $[K : F] = |G|$ is finite, so $|\text{Aut}(K/F)| \leq [K : F] = |G|$. Since $G \leq \text{Aut}(K/F)$, it follows $G = \text{Aut}(K/F)$ and K/F is Galois.

Corollary: If $G_1 \neq G_2 \leq \text{Aut}(K)$ are finite, then $K^{G_1} \neq K^{G_2}$.

Proof: Suppose $K^{G_1} = K^{G_2}$. Then $\text{Aut}(K : K^{G_1}) = \text{Aut}(K : K^{G_2})$, so $G_1 = G_2$.

Claim: Suppose K/F is Galois. Then K is the splitting field of a separable polynomial over F .

Proof: Let $G = \text{Aut}(K/F)$ and pick $\{\alpha_{ij}\}$ such that $\{\alpha_{ij}\}$ for fixed i and variable j is the G -orbit of α_{i1} , the orbits $\{\alpha_{ij}\}$ for variable j are disjoint, and that $\{\alpha_{ij}\}$ generates K/F . Let $f = \prod_{ij} (x - \alpha_{ij})$. Note that $\sigma f = f$ for all $\sigma \in G$, so the coefficients of f are in $K^G = F$.

f is separable over F and splits over E , so we're done.

Claim: Further, if the above holds, then every irreducible in F which has a root in K is separable and has all roots in K .

Proof: Let $f \in F[x]$ be an irreducible and $\alpha \in K$ be a root. Then each element of the orbit $G\alpha$ is a root of f and $\prod_{\beta \in G\alpha} (x - \beta) \in F[x]$ is a divisor of f , so $f = \prod_{\beta \in G\alpha} (x - \beta)$.

Let S denote the set of subfields L such that $K/L/F$. Let T denote the set of subgroups of $\text{Aut}(K/F)$. Define $\varphi : S \rightarrow T$ by $\varphi(L) = \text{Aut}(K/L)$. For $H \leq \text{Aut}(K/F)$, define $\psi(H) = K^H$.

Note that $\varphi\psi(H) = \varphi(K^H)$. We have $\text{Aut}(K : K^H) \geq H$ and $[K : K^H] = |H|$. Since $|\text{Aut}(K : K^H)| \leq [K : K^H] = |H|$, we have $\text{Aut}(K : K^H) = H$. In other words, $\varphi\psi(H) = H$ always when K/F is finite.

Now we wish to show $\psi\varphi(L) = L$ or $K^{\text{Aut}(K/L)} = L$.

Claim: K/F is Galois iff $F = K^{\text{Aut}(K/F)}$.

Proof: Suppose $F = K^{\text{Aut}(K/F)}$. Then $[K : F] = [K : K^{\text{Aut}(K/F)}] = |\text{Aut}(K/F)|$. Conversely, suppose K/F is Galois. Then $[K : K^{\text{Aut}(K/F)}] = |\text{Aut}(K/F)| = [K : F]$. Since $F \leq K^{\text{Aut}(K/F)}$, we have $F = K^{\text{Aut}(K/F)}$.

Let K/F be Galois. Then K is the splitting field of some separable $f \in F[x]$. K is the splitting field of $f \in L[x]$ as well, so K/L is Galois. Hence, $L = K^{\text{Aut}(K/L)}$.

(1). If E_1, E_2 correspond to H_1, H_2 , then $E_1 \leq E_2$ iff $H_1 \geq H_2$. Proof: Suppose $E_1 \leq E_2$. Then $\text{Aut}(K/E_1) \geq \text{Aut}(K/E_2)$, ie $H_1 \geq H_2$. Conversely, if $H_1 \geq H_2$, then $K^{H_1} \leq K^{H_2}$, ie $E_1 \leq E_2$.

(2). Let E correspond to H . Then $[E : F] = [G : H]$. Proof: $[E : F] = [K : F]/[K : E] = |\text{Aut}(K/F)|/|\text{Aut}(K/E)| = |G|/|H| = [G : H]$.

(3). K/E is Galois with Galois group H .

(4). E/F is Galois iff $H \trianglelefteq G$. Proof: Suppose E/F is Galois. Let $\varphi : G \rightarrow \text{Aut}(E/F)$ be the restriction of $\sigma \in G$ to $\sigma|_K$. I claim that φ is well-defined. Indeed, E is the splitting field of some separable $f \in F[x]$. Let f have roots $\alpha_1, \alpha_2, \dots, \alpha_n \in E$. Then $\sigma(\alpha_i)$ is sent to $\sigma(\alpha_j)$, so $\sigma(K) = K$.

I claim that $\ker \varphi = H$. Suppose $\sigma|_E = 1$. Then σ fixes E . It follows that $H \trianglelefteq G$ by the First Isomorphism theorem. I claim that φ is surjective. Indeed, let $\tau \in \text{Aut}(E/F)$. K is the splitting field of some separable $g \in F[x] \leq E[x]$. Then τ extends to some $\tilde{\tau} \in G$. It follows by the First Isomorphism theorem that $\text{Aut}(E/F) \cong G/H$.

Let $\overline{F}$ be an algebraic closure of F containing K . Let $\sigma \in \text{Aut}(K/F)$. Since E/F is finite, so is $\sigma(E)/F$ (with the same index), so $\sigma(E) \leq \overline{F}$. Conversely, suppose $\tau : E \rightarrow \overline{F}$ is an embedding fixing F . I claim that $\tau(E) \leq K$. Let

$\alpha \in E$. Then τ fixes $m_{\alpha,F}$. Since $m_{\alpha,F}$ has a root in K , $m_{\alpha,F}$ splits over K . τ sends α to another root β of $m_{\alpha,F}$, so $\tau(\alpha) = \beta \in K$. Every embedding of E is of the form $\sigma|_E$ for some $\sigma \in G$.

Let $\sigma_1, \sigma_2 \in G$ and suppose $\sigma_1|_E = \sigma_2|_E$. Then $\sigma_1\sigma_2^{-1}|_E = 1$, so $\sigma_1H = \sigma_2H$. The number of embeddings of E/F is $[G : H] = [E : F]$.

Let $H \leq G$ and $\sigma \in G$. I claim that $\text{Aut}(K/\sigma(E)) = \sigma\text{Aut}(K/E)\sigma^{-1}$. It suffices to show $\text{Aut}(K/\sigma(E)) \leq \sigma\text{Aut}(K/E)\sigma^{-1}$ since we would have $\text{Aut}(K/E) \leq \sigma^{-1}\text{Aut}(K/\sigma(E))\sigma$. Let $\tau \in \text{Aut}(K/\sigma(E))$. Then $\sigma^{-1}\tau\sigma \in \text{Aut}(K/E)$, and we're done.

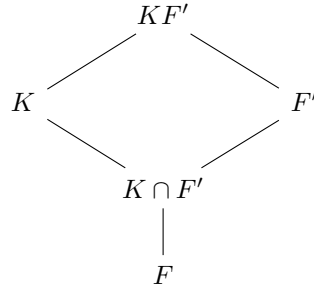
We have $\text{Aut}(K/\sigma(E)) = \text{Aut}(K/E)$. Since K/F is Galois, we have $\sigma(E) = E$ by the correspondence.

Let $(X, \leq)$ be a partially ordered set with the property that if $x_1, x_2 \in X$, there exists $x_3 \leq x_1, x_2$ such that if $x \leq x_1, x_2$, then $x \leq x_3$. Let there be a similar property for the upper bounds. Suppose x, x' have this property. Since $x, x' \leq x_1, x_2$, we have $x \leq x'$ and $x' \leq x$. Then $x = x'$. Denote by $m(x_1, x_2)$ this lower number (the “min”) and $M(x_1, x_2)$ the max).

Let $\varphi : (X, \leq) \rightarrow (Y, \leq)$ be order preserving and surjective. Then $\varphi(m(x_1, x_2)) = m(\varphi(x_1), \varphi(x_2))$.

Proof: $m(x_1, x_2) \leq x_1, x_2$ and if $x \leq x_1, x_2$, then $x \leq m(x_1, x_2)$. Then $\varphi(m(x_1, x_2)) \leq \varphi(x_1), \varphi(x_2)$ (by order preservation), and if $\varphi(x) \leq \varphi(x_1), \varphi(x_2)$, then $\varphi(x) \leq \varphi(m(x_1, x_2))$. Since φ is surjective, $\varphi(x)$ can be replaced by $y \in Y$. It follows $\varphi(m(x_1, x_2)) = m(\varphi(x_1), \varphi(x_2))$. Similarly, $\varphi(M(x_1, x_2)) = M(\varphi(x_1), \varphi(x_2))$. If φ is a contravariant map (as with the Galois correspondence), I think the same can be done.

3 Composite Extensions and Simple Extensions



Suppose K/F is Galois. Let $f \in F[x]$ be separable and E/F be the splitting field of f . $f \in F'[x]$ splits over KF' . Suppose L/F' such that f splits. Then $L/K, F'$, so $L \geq KF'$, so KF' is the splitting field of $f \in F'[x]$.

Consider the map $\text{Aut}(KF'/F') \rightarrow \text{Aut}(K/K \cap F')$ given by $\sigma \mapsto \sigma|_K$. This map is well-defined since $\sigma(K) = K$ (K/F is splitting and F is fixed) and $K \cap F'$ is fixed by σ . The map is trivially a homomorphism.

Let H denote the image of the map. K^H, F' are fixed by $\text{Aut}(KF'/F')$, so $K^H F'$ is fixed by $\text{Aut}(KF'/F')$. It follows $K^H F' = F'$, so $K^H \leq F'$. Then $K^H \leq K \cap F'$, so $K^H = K \cap F'$, so $H = \text{Aut}(K/K \cap F')$, as desired.

Corollary:

$$\begin{aligned} [KF' : F] &= [KF' : F'] [F' : F] = [K : K \cap F'] [F' : F] \\ &= \frac{[K : F] [F' : F]}{[K \cap F' : F]}. \end{aligned}$$

Note that the above equation is symmetric with respect to K, F' , though only at least one of K, F' are required to be Galois over F . Take $F = \mathbb{Q}, K = \mathbb{Q}(\sqrt[3]{2}), F' = \mathbb{Q}(\rho\sqrt[3]{2})$. Compute $KF' = \mathbb{Q}(\sqrt[3]{2}, \rho)$ and $K \cap F' = \mathbb{Q}$. The RHS of the formula is $3 \cdot 3/1 = 9$, but the LHS is 9.

Let $K_1, K_2/F$ be Galois. Let $\{\alpha_i\}_{i \in [n]}$ generate $K_1 \cap K_2/F$. Since K_1/F is Galois, $m_{\alpha_i, F}$ splits into distinct roots over K_1 . Similarly, $m_{\alpha_i, F}$ splits into distinct roots over K_2 . It follows that $m_{\alpha_i, F}$ splits into distinct roots over $K_1 \cap K_2$. Then $E_1 \cap E_2$ is the splitting field of the separable part of the product of the $m_{\alpha_i, F}$.

I claim that $K_1 K_2/F$ is Galois. Indeed, let $K_1, K_2/F$ be the splitting fields of the separables $f_1, f_2 \in F[x]$. Then $K_1 K_2$ is the splitting field of the separable part of $f_1 f_2$.

$\text{Aut}(K_1 K_2/F) \rightarrow \text{Aut}(K_1/F) \times \text{Aut}(K_2/F)$ is injective. Suppose $\sigma \mapsto (\sigma_1, \sigma_2)$. Then $\sigma_1|_{(K_1 \cap K_2)} = \sigma_2|_{(K_1 \cap K_2)}$. Denote by $H = \{(\sigma_1, \sigma_2) \in \text{Aut}(K_1/F) \times \text{Aut}(K_2/F) : \sigma_1|_{K_1 \cap K_2} = \sigma_2|_{K_1 \cap K_2}\}$. We compute $|H|$ as follows. $K_1, K_2/K_1 \cap K_2$ are Galois, so given $\sigma \in \text{Aut}(K_1 \cap K_2/F)$, there are exactly $[K_1 : K_1 \cap K_2], [K_2 : K_1 \cap K_2]$ extensions of σ to K_1, K_2 . Hence,

$$\begin{aligned} |H| &= [K_1 : K_1 \cap K_2] \cdot [K_2 : K_1 \cap K_2] \cdot |\text{Aut}(K_1 \cap K_2/F)| \\ &\leq \frac{[K_1 : F] [K_2 : F]}{[K_1 \cap K_2 : F]} = [K_1 K_2 : F], \end{aligned}$$

so the map is an isomorphism. Further, equality holds, so $|\text{Aut}(K_1 \cap K_2/F)| = [K_1 \cap K_2 : F]$.

If $K_1 \cap K_2 = F$, then $H = \text{Aut}(K_1/F) \times \text{Aut}(K_2/F)$. Suppose for $G = \text{Aut}(K/F)$ that $G \cong G_1 \times G_2$ for $G_1, G_2 \leq G$. Let $K_1 = K^{G_1}, K_2 = K^{G_2}$. Then $K_1 \cap K_2 = K^{\langle G_1, G_2 \rangle} = K^G = F$ and $K_1 K_2 = K^{G_1 \cap G_2} = K^1 = F$.

Let $\{V_\alpha\}$ be a collection of subspaces. Then $\bigcap_\alpha V_\alpha = \bigcap_{i \in [n]} V_{\alpha_i}$ for some finite indexing set $[n]$.

Proof: Pick $\alpha_1 \in J$ to be arbitrary. Define $W_1 = V_{\alpha_1}$. Suppose W_n is defined. If $W_n \leq V_\alpha$ for all $\alpha \in J$, then we're done since $\bigcap_{\alpha \in J} V_\alpha \leq W_n$. Else, $V_\alpha \cap W_n < W_n$. Let $\alpha_{n+1} = \alpha$ and $W_{n+1} = W_n \cap V_{\alpha_{n+1}}$. Suppose FT SOC W_n is defined for all $n \in \mathbb{Z}^+$. Then the set $\{\dim W_n : n \in \mathbb{Z}^+\}$ is a strictly descending set of nonnegative integers since $W_{n+1} < W_n$ for all n , absurd.

Let E/F be finite dimensional separable. Let $\overline{F}$ contain E . Let $\{\alpha_i\}_{i \in [n]}$ generate E/F . Let $\overline{F}/K/F$ be the splitting field of the separable part of the product of the $m_{\alpha_i, F}$, so K is Galois.

Then define $\text{Gal}(E) = \bigcap_{E \leq L \leq \overline{F}: L/F \text{ is Galois}} L$, which is well-defined since the indexing set is nonempty. This L/F is a finite-dimensional vector space, so $\text{Gal}(E)$ is a finite intersection of Galois extensions of F , and therefore Galois.

Let $\overline{F}_1, \overline{F}_2/E$ be algebraic closures of F , where E/F is finite separable. Let $K_1, K_2/F$ denote the Galois closures of E inside $\overline{F}_1, \overline{F}_2$.

Claim: Let $\overline{F}/F$ denote an algebraic closure and assume $\overline{F}/E/F$. Then $\overline{E} = \overline{F}$.

Proof: Since $E \leq \overline{F}$, we have $\overline{E} \leq \overline{\overline{F}} = \overline{F}$. Further, $\overline{E} \geq \overline{F}$, so $\overline{E} = \overline{F}$.

Let $\varphi : \overline{F}_1 \rightarrow \overline{F}_2$ denote an isomorphism fixing E . If L/E is Galois, then $\varphi(L)/E$ is also Galois. Then $\varphi(K_1) = K_2$.

Let K/F be finite. Then $K = F(\theta)$ iff there exist finitely many subfields of K containing F .

Suppose $K = F(\theta)$ and $K/E/F$. Let f, g denote the minimal polynomials of θ over F, E , so $g \mid f$. Let $E' \leq E$ denote the subfield generated by the coefficients of g . Note that the minimal polynomial of θ over E, E' is g in both instances, so $[K : E] = [K : E'] = \deg g$, so $E = E'$. It follows that we can identify the subfields E with the divisors of f in $E[x]$, of which there are finitely many.

Now suppose that there are finitely many subfields of K/F . If F is finite, we're done. Assume F is infinite. Since K/F is finitely generated, it suffices to show that $F(\alpha, \beta)$ is simple.

Consider the fields $F(\alpha + a\beta)$ for $a \in F$. Note that $F \leq F(\alpha + a\beta) \leq F(\alpha, \beta)$. Since F is infinite and there are finitely many subfields of K/F , there exist $a, a' \in F$ such that $F(\alpha + a\beta) = F(\alpha + a'\beta) = L$. Then $\alpha + a\beta, \alpha + a'\beta \in L$, so $(a - a')\beta \in L$, so $\beta \in L$. Then $\alpha \in L$, so $L = F(\alpha, \beta)$.

Primitive Element theorem. If K/F is finite and separable, then K/F is simple.

Proof: Since K/F is finite separable, K has a Galois closure $\tilde{K}$. $\text{Aut}(\tilde{K}/F)$ has $[\tilde{K} : F]$ many elements, so $\text{Aut}(\tilde{K}/F)$ has finitely many subgroups. By the Galois correspondence, there are finitely many subfields of $\tilde{K}/F$. Hence, K/F is simple.

4 Cyclotomic Extensions

Claim: $\text{Aut}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong \mathbb{Z}_n^\times$.

Proof: Define $\Phi : \mathbb{Z}_n^\times \rightarrow \text{Aut}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ by $\Phi(a) = \sigma_a$, which is well-defined since $\sigma_a = \sigma_{a'}$ for $n \mid a - a'$. Φ is injective and the domain, codomain are finite with the same cardinality. Hence, Φ is bijective. Finally,

$$\Phi(ab)(\zeta_n) = \sigma_{ab}(\zeta_n) = \zeta_n^{ab} = (\zeta_n^a)^b = \sigma_a(\zeta_n^b) = \sigma_a \sigma_b(\zeta_n) = \Phi(a)\Phi(b)(\zeta_n),$$

so Φ is indeed a homomorphism.

Let $H \leq \text{Aut}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$ and define $\alpha = \sum_{\sigma \in H} \sigma \zeta_p$, so α is fixed pointwise by H . Note that $\{\zeta_p, \zeta_p^2, \dots, \zeta_p^{n-1}\}$ is a basis of $\mathbb{Q}(\zeta_p)/\mathbb{Q}$ and these elements are permuted by $\text{Aut}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$. An automorphism $\tau \in \text{Aut}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$ is determined by $\tau(\zeta_p)$. Hence, if $A, B \leq \text{Aut}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$ are nonempty disjoint, then $\sum_{\sigma \in A} \sigma(\zeta_p) \neq \sum_{\sigma \in B} \sigma(\zeta_p)$. In particular, cosets are disjoint, so if $\tau \notin H$, then $\sum_{\sigma \in H} \tau \sigma(\zeta_p) \neq \sum_{\sigma \in H} \sigma(\zeta_p)$, ie $\tau(\alpha) \neq \alpha$. The group of automorphisms fixing α is H , so by the Galois correspondence, $\mathbb{Q}(\alpha) = \mathbb{Q}(\zeta_p)^H$.

Claim: Write $n = \prod_{i \in [k]} p_i^{a_i}$. Then $\mathbb{Q}(\zeta_n) = \mathbb{Q}(\zeta_{p_1^{a_1}}) \mathbb{Q}(\zeta_{p_2^{a_2}}) \dots \mathbb{Q}(\zeta_{p_k^{a_k}})$.

Proof: Take $\zeta_{p_i^{a_i}} = \zeta_n^{\prod_{j \neq i} p_j^{a_j}}$, so $\zeta_{p_i^{a_i}} \in \mathbb{Q}(\zeta_n)$. It follows that the composite of the $\mathbb{Q}(\zeta_{p_i^{a_i}})$ is contained in $\mathbb{Q}(\zeta_n)$. Conversely,

$$\prod_{i \in [k]} \left(\zeta_n^{\prod_{j \neq i} p_j^{a_j}} \right) = \zeta_n^{\sum_{i \in [k]} \prod_{j \neq i} p_j^{a_j}}.$$

Denote the exponent of ζ_n by N . Note that

$$\gcd(N, p_i) = \gcd \left(\prod_{j \neq i} p_j^{a_j}, p_i \right) = 1,$$

so $\gcd(N, n) = 1$. It follows that ζ_n^N is a primitive n th root of unity.

Claim: Assume all fields in question are contained in a common field. Let K_i/F be a finite extension and $K = \bigvee_{i \in [n]} K_i$ such that $[K : F] = \prod_{i \in [n]} [K_i : F]$. Then for any disjoint subsets $A_1, A_2 \leq [n]$ and $E_1 = \bigvee_{i \in A_1} K_i$, $E_2 = \bigvee_{i \in A_2} K_i$, that $E_1 \cap E_2 = F$. If $A_i = \emptyset$, then treat $E_i = F$.

Proof: Disregard the claim's notation. Let $K_1, K_2/F$. It suffices to show if $K_1 \cap K_2 > F$, then $[K_1 K_2 : F] < [K_1 : F] \cdot [K_2 : F]$. Let $\{\alpha_i\}_{i \in [n]}$ and $\{\beta_j\}_{j \in [m]}$ be bases of K_1, K_2 over F . Then $K_1 = F(\alpha_i : i \in [n])$ and similarly for K_2 . It follows that $K_1 K_2 = F(\alpha_i, \beta_j : i \in [n], j \in [m]) = F[\alpha_i, \beta_j : i \in [n], j \in [m]]$, where the second equality follows from the α_i, β_j being algebraic over F .

I claim that $\{\alpha_i \beta_j : i \in [n], j \in [m]\}$ generates $K_1 K_2/F$. Indeed, $K_1 K_2$ can be treated as the set of multivariate polynomials in α_i, β_j over F . If $\alpha_{i_1}, \alpha_{i_2}$ are multiplied in a term, since the $\{\alpha_i\}$ is a basis of K_1 , we can replace $\alpha_{i_1} \alpha_{i_2}$ with a F -linear combination of the α_i . Repeating the argument gives that $K_1 K_2$ is spanned by the $\alpha_i \beta_j$.

The choice of the bases of $K_1, K_2/F$ were arbitrary. Since independent sets can be extended, WLOG $\alpha_1 = \beta_1 = 1$. Suppose $x \in K_1 \cap K_2 - F$. Write $x = \sum_{i \in [n]} x_i \alpha_i = \sum_{j \in [m]} x'_j \beta_j$, so we have

$$\sum_{i \in [n]} x_i \alpha_i \beta_1 = \sum_{j \in [m]} x'_j \alpha_1 \beta_j.$$

Note that $x_i \neq 0$ for some $i \neq 1$ since $x \notin F$. WLOG $x_2 \neq 0$. The LHS contains a nonzero summand of $\alpha_2 \beta_1$, while the RHS doesn't, so the set $\{\alpha_i \beta_j : i \in [n], j \in [m]\}$ is not independent. It follows that $[K_1 K_2 : F] < [K_1 : F] \cdot [K_2 : F]$.

We finish the above proof. If there exist E_1, E_2 such that $E_1 \cap E_2 \neq F$, then $[E_1 E_2 : F] < [E_1 : F] \cdot [E_2 : F]$. Then $K = \left(\bigvee_{i \in [n] - (A_1 \cup A_2)} K_i \right) \vee E_1 E_2$, and

$$\begin{aligned} [K : F] &\leq \prod_{i \in [n] - (A_1 \cup A_2)} [K_i : F] \cdot [E_1 E_2 : F] < \left(\prod_{i \in [n] - (A_1 \cup A_2)} [K_i : F] \right) \cdot [E_1 : F] \cdot [E_2 : F] \\ &\leq \left(\prod_{i \in [n] - (A_1 \cup A_2)} [K_i : F] \right) \cdot \prod_{i \in A_1} [K_i : F] \cdot \prod_{i \in A_2} [K_i : F] = \prod_{i \in [n]} [K_i : F], \end{aligned}$$

absurd.

Since $[\mathbb{Q}(\zeta_{p_i^{a_i}}) : \mathbb{Q}] = \varphi(p_i^{a_i})$, $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$, and that φ is multiplicative, by the above result, we have that the $\mathbb{Q}(\zeta_{p_i^{a_i}})$ are mutually "independent."

Claim: Let Galois extensions K_i/F be mutually independent for $i \in [n]$. Let

$K = \bigvee_{i \in [n]} K_i$, so K/F is Galois. Then $\text{Aut}(K/F) \cong \prod_{i \in [n]} \text{Aut}(K_i/F)$.

Proof: Since K_i/F is Galois, K_i is the splitting field of a separable polynomial $f_i \in F[x]$. Hence, if $\sigma \in \text{Aut}(K/F)$, then $\sigma|_{K_i} \in \text{Aut}(K_i/F)$. We have a homomorphism

$$\Phi : \text{Aut}(K/F) \rightarrow \prod_{i \in [n]} \text{Aut}(K_i/F).$$

I claim that Φ is injective. If $\sigma|_{K_i} = 1$ for all $i \in [n]$, then K_i is contained in the fixed field K^σ of σ for all i . Hence, $K = \bigvee_{i \in [n]} K_i$ is contained in K^σ , so σ fixes K . By order considerations, we have that Φ is bijective and hence an isomorphism.

As an application, we take $K_i = \mathbb{Q}(\zeta_{p_i^{a_i}})$. Under the isomorphism $\text{Aut}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong \mathbb{Z}_n^\times$, we have the Chinese Remainder theorem.

5 Galois Groups of Polynomials

Let x_i be an indeterminate for $i \in [n]$. Let s_i be the i th symmetric sum for $i \in [n]$. Let $f = \prod_{i \in [n]} (x - x_i)$ be the general polynomial of degree n , so $f \in F(s_i : i \in [n])$. Note that the splitting field of f is $F(x_i : i \in [n])$, so $[F(x_i : i \in [n]) : F(s_i : i \in [n])] \leq n!$.

Conversely, we have an inclusion $S_n \hookrightarrow \text{Aut}(F[x_i : i \in [n]])$ and therefore an inclusion $S_n \hookrightarrow \text{Aut}(F(x_i : i \in [n]))$. The elements of S_n fix each s_i , so we in fact have the inclusion $S_n \hookrightarrow \text{Aut}(F(x_i : i \in [n])/F(s_i : i \in [n]))$. It follows that $S_n \rightarrow \text{Aut}(F(x_i : i \in [n])/F(s_i : i \in [n]))$ is an isomorphism, and that $F(x_i : i \in [n])/F(s_i : i \in [n])$ is Galois with group S_n .

It follows that if $r \in F(x_i : i \in [n])$ is a rational function of the x_i that is fixed by each element of S_n , then $r \in F(s_i : i \in [n])$.

Discriminant of the Cubic. Let $f = x^3 + ax^2 + bx + c$. Make the substitution $x \mapsto x - \frac{a}{3}$, so f transforms to $f = x^3 + px + q$. Let f have roots $\alpha_1, \alpha_2, \alpha_3$. Note

$$\begin{aligned} f'(\alpha_1) &= (\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3) \\ f'(\alpha_2) &= (\alpha_2 - \alpha_1)(\alpha_2 - \alpha_3) \\ f'(\alpha_3) &= (\alpha_3 - \alpha_1)(\alpha_3 - \alpha_2) \end{aligned}$$

so

$$\begin{aligned}
D &= -f'(\alpha_1)f'(\alpha_2)f'(\alpha_3) = -(3\alpha_1^2 + p)(3\alpha_2^2 + p)(3\alpha_3^2 + p) \\
&= 27(\sqrt{-p/3} - \alpha_1)(\sqrt{-p/3} - \alpha_2)(\sqrt{-p/3} - \alpha_3)(\sqrt{-p/3} + \alpha_1)(\sqrt{-p/3} + \alpha_2)(\sqrt{-p/3} + \alpha_3) \\
&= -27f(\sqrt{-p/3})f(-\sqrt{-p/3}) \\
&= -27((-p/3)^{3/2} + p(-p/3)^{1/2} + q)(-(-p/3)^{3/2} - p(-p/3)^{1/2} + q) \\
&= -27\left(\frac{2p}{3}\sqrt{-p/3} + q\right)\left(-\frac{2p}{3}\sqrt{-p/3} + q\right) \\
&= -27\left(q^2 - \frac{4p^2}{9}(-p/3)\right) \\
&= -4p^3 - 27q^2.
\end{aligned}$$

Note that f becomes

$$\begin{aligned}
g &= \left(x - \frac{a}{3}\right)^3 + a\left(x - \frac{a}{3}\right)^2 + b\left(x - \frac{a}{3}\right) + c \\
&= \left(x^3 - ax^2 + \frac{a^2}{3}x - \frac{a^3}{27}\right) + a\left(x^2 - \frac{2a}{3}x + \frac{a^2}{9}\right) + b\left(x - \frac{a}{3}\right) + c \\
&= x^3 + \left(b - \frac{a^2}{3}\right) + \left(\frac{2}{27}a^3 - \frac{ab}{3} + c\right),
\end{aligned}$$

so $p = b - \frac{a^2}{3}$ and $q = \frac{2}{27}a^3 - \frac{ab}{3} + c$ and

$$\begin{aligned}
D &= -4\left(b - \frac{a^2}{3}\right)^3 - 27\left(\frac{2}{27}a^3 - \frac{ab}{3} + c\right)^2 \\
&= -4\left(b^3 - a^2b^2 + \frac{a^4b}{3} - \frac{a^6}{27}\right) - 27\left(\frac{4}{729}a^6 + \frac{a^2b^2}{9} + c^2 - \frac{4a^4b}{81} + \frac{4a^3c}{27} - \frac{2abc}{3}\right) \\
&= -4b^3 + 4a^2b^2 - \frac{4a^4b}{3} + \frac{4a^6}{27} - \frac{4a^6}{27} - 3a^2b^2 - 27c^2 + \frac{4a^4b}{3} - 4a^3c + 18abc \\
&= -4b^3 + a^2b^2 - 27c^2 - 4a^3c + 18abc.
\end{aligned}$$

Note for $a = 0$, we have $D = -4b^3 - 27c^2$, as expected.

Let p be an odd prime. Then the discriminant of ϕ_p is $(-1)^{(p-1)/2}p^{p-2}$.

Proof: Let ζ be a primitive p th root of unity. Write symmetrically

$$D = (-1)^{(p-1)p/2} \prod_{i \in [p-1]} \prod_{j \neq i} (\zeta^i - \zeta^j) = (-1)^{(p-1)p/2} \prod_{i \in [p-1]} \phi'_p(\zeta^i).$$

We have $\phi_p(x) = \frac{x^p-1}{x-1}$, so $\phi'_p(x) = \frac{px^{p-1}(x-1)-(x^p-1)}{(x-1)^2}$. If x is a p th root of unity, then $\phi'_p(x) = \frac{p}{x(x-1)}$. Then

$$\begin{aligned} D &= (-1)^{(p-1)/2} \prod_{i \in [p-1]} \frac{p}{\zeta^i(\zeta^i-1)} = (-1)^{(p-1)/2} p^{p-1} (-1)^{(p-1)} (-1)^{(p-1)} / \phi_p(1) \\ &= (-1)^{(p-1)/2} p^{p-2}, \end{aligned}$$

as desired. In particular, $\mathbb{Q}(\sqrt{(-1)^{(p-1)/2}p}) \leq \mathbb{Q}(\zeta_p)$.

Claim: Fundamental Theorem of Algebra.

Proof: Let f be a real polynomial of degree $n > 0$. Write $n = 2^k r$, where $2 \nmid r$. We induct on k . If $k = 0$, then f has odd degree and we're done by IVT. Else, let f have roots α_i for $i \in [n]$ in some splitting field R over $\mathbb{R}$ (which is Galois since $\text{char } \mathbb{R} = 0$). Consider for each $t \in \mathbb{R}$:

$$L_t = \prod_{1 \leq i < j \leq n} (x - \alpha_i - \alpha_j + t\alpha_i\alpha_j),$$

which is invariant under permutations of the roots α_i . Hence $L_t \in \mathbb{R}[x]$.

Note that L_t has degree $n(n-1)/2 = 2^{k-1}r(2^k r - 1)$, where $r(2^k r - 1)$ is odd. By the IH, L_t has a complex root, which is equal to $\alpha_i + \alpha_j - t\alpha_i\alpha_j$ for some $1 \leq i < j \leq n$. Since this holds for each $t \in \mathbb{R}$ and $\mathbb{R}$ is infinite, by Pigeon-hole principle, WLOG $\alpha_1 + \alpha_2 - t\alpha_1\alpha_2, \alpha_1 + \alpha_2 - s\alpha_1\alpha_2 \in \mathbb{C}$ for distinct $s, t \in \mathbb{C}$.

It follows that $S = \alpha_1 + \alpha_2, P = \alpha_1\alpha_2 \in \mathbb{C}$. α_1, α_2 are the roots of $x^2 - Sx + P \in \mathbb{C}[x]$, which we determined to have roots in $\mathbb{C}$. Hence, $\alpha_1, \alpha_2 \in \mathbb{C}$.

6 Solvable and Radical Extensions: Insolubility of the Quintic

Claim: Let $\text{char } F \nmid n$ and E/F be the splitting field of $f = x^n - 1$. Then the roots of f are generated by a primitive element ζ .

Proof: The n -Torsion of F is cyclic of order at most n . Since $\text{char } F \nmid n$, f is separable and the n -Torsion is $\mathbb{Z}_n$.

Claim: Let $a \in F$. Then $F(\sqrt[n]{a})/F$ is cyclic of order dividing n .

Proof: Let ζ be a primitive n th root of unity. Let $m \sqrt[n]{a}, F$ have root $\zeta^g \sqrt[n]{a}$, where $g \in \mathbb{Z}^+$ is minimal (which exists since $\zeta^n = 1$). There exists an isomorphism $\sigma_g : F(\sqrt[n]{a}) \rightarrow F(\zeta^g \sqrt[n]{a}) = F(\sqrt[n]{a})$ such that $\sqrt[n]{a} \mapsto \zeta^g \sqrt[n]{a}$. Then we

have the automorphisms $\sigma_{kg} = \sigma_g^k$, where $\sqrt[n]{a} \mapsto \zeta^{kg} \sqrt[n]{a}$ for all $k \in \mathbb{Z}$. Suppose FTSOC $g \nmid n$. Write by the Division algorithm $n = gq + r$, where $0 < r < g$, so $\sigma_r = \sigma_0 \sigma_{g(-q)}$. It follows $\sigma_r(\sqrt[n]{a}) = \zeta^r \sqrt[n]{a}$ is a root of $m_{\sqrt[n]{a}, F}$, absurd.

Since $F(\sqrt[n]{a})/F$ is Galois, $m_{\sqrt[n]{a}, F}$ contains the roots $\sigma_{kg}(\sqrt[n]{a}) = \zeta^{kg} \sqrt[n]{a}$. If $m_{\alpha, F}$ contains another root $\zeta^m \sqrt[n]{a}$, write $m = gq + r$ and apply the previous argument. Hence, we may write $m_{\sqrt[n]{a}, F} = \prod_{i \in [n/g]} (x - \zeta^{ig} \sqrt[n]{a})$. $F(\sqrt[n]{a})/F$ has degree $\deg m_{\sqrt[n]{a}, F} = n/g$, which divides n and Galois group $\mathbb{Z}_{n/g}$.

We have a map $\text{Aut}(F(\sqrt[n]{a})/F) \rightarrow U_n$, where $\sigma \mapsto \zeta_\sigma$, where $\sigma(\sqrt[n]{a}) = \zeta_\sigma \sqrt[n]{a}$. This map is a homomorphism:

$$\sigma\tau(\sqrt[n]{a}) = \sigma(\zeta_\tau \sqrt[n]{a}) = \zeta_\tau \sigma(\sqrt[n]{a}) = \zeta_\tau \zeta_\sigma \sqrt[n]{a} = \zeta_\sigma \zeta_\tau \sqrt[n]{a}$$

(so $\zeta_{\sigma\tau} = \zeta_\sigma \zeta_\tau$).

This map is injective since the image of σ on $\sqrt[n]{a}$ determines σ . Hence, $\text{Aut}(F(\sqrt[n]{a})/F)$ is embedded as a subgroup of U_n , and therefore cyclic of order dividing n .

Let K/F be a cyclic extension of degree n over a field containing the n th roots of unity not of characteristic dividing n . Let σ be a generator of $\text{Aut}(K/F)$. For $\alpha \in K$ and any n th root of unity ζ , define the Lagrange resolvent $(\alpha, \zeta) \in K$ by $(\alpha, \zeta) = \sum_{i=0}^{n-1} \zeta^i \sigma^i(\alpha)$.

I claim that (α, ζ) is independent of σ . Note that

$$\begin{aligned} \sigma(\alpha, \zeta) &= \sum_{i=0}^{n-1} \zeta^i \sigma^{i+1}(\alpha) = \zeta^{-1} \sum_{i=0}^{n-1} \zeta^{i+1} \sigma^{i+1}(\alpha) = \zeta^{-1} \sum_{i=1}^n \zeta^i \sigma^i(\alpha) \\ &= \zeta^{-1} \sum_{i=0}^{n-1} \zeta^i \sigma^i(\alpha) = \zeta^{-1} (\alpha, \zeta). \end{aligned}$$

It follows that $\sigma(\alpha, \zeta)^n = \zeta^{-n} (\alpha, \zeta)^n = (\alpha, \zeta)^n$, so $(\alpha, \zeta)^n \in F$. The automorphisms σ^i are distinct for $i \in [n-1]$. By Linear Independence of characters, there exists $\alpha \in K$ such that $(\alpha, \zeta) \neq 0$. We have $\sigma^i(\alpha, \zeta) = \zeta^{-i} (\alpha, \zeta)$ for all $i \in [n-1]$, so σ^i doesn't fix (α, ζ) for all i . Hence, (α, ζ) can't lie in any proper subfield of K . It follows that $F((\alpha, \zeta)) = F(\sqrt[n]{a})$, where $a = (\alpha, \zeta)^n$.